Appendix

The following tables provide supplementary data referenced in the main text.

Table S1. CaptchaShield Dataset Statistics (Illusion Diffusion Pipeline). The full benchmark comprises 420,000 base images; a stratified subset of 1,000 images is used for attack experiments and VLM assessment.

Component	**Count**
Chinese characters	3,500
Background images	120
Base images total	420,000
Evaluation subset (stratified)	
Source images	1,000
Attacked images (×6 methods)	6,000
VLM API calls (Q1+Q2+Q3×3 VLMs)	54,000
Grand Total (base)	**420,000**

Table S2. Adversarial Perturbation Methods Benchmarked in CaptchaShield. *Category* indicates the primary research community origin. *Modality* denotes the reference signal used to guide optimization.

Method	**Venue**	**Category**	**Modality**
ASPL [15]	ICCV 2023	Fine-tune disruption	Image-only
Glaze [14]	USENIX Sec 2023	Style protection	Image-only
AMP [12]	NeurIPS 2023	VLM transfer attack	Image-only
XTransfer [17]	ICLR 2024	Transfer enhancement	Image-only
Nightshade [13]	IEEE S&P 2024	T2I data poisoning	Text-only
MMCoA [16]	arXiv 2024	Multimodal joint	Image+Text

Table S3. VLM API Versions and Evaluation Call Window. All API calls used `max_tokens`=64 and default temperature. Calls were issued sequentially per image across three questions in parallel threads; timeout was set to 10 s per call.

VLM	Model / Deployment ID	API Version	Call Window
GPT-5.2	gpt-5.2 (Azure OpenAI)	2024-12-01-preview	Jan–Mar 2026
Gemini 3.0	gemini-3-flash-preview (Google)	v1beta (genai SDK)	Jan–Mar 2026
GLM-4V	glm-4v-flash (Zhipu AI)	v4 (OpenAI-compat.)	Jan–Mar 2026

Table S4. Evaluation Question Design (Q1–Q3). Each attacked image is probed with three questions measuring different dimensions of adversarial protection.

Q	Task (VLM prompt)	Metric	Goal
Q1	Three-way forced choice: source char, target char, or neither	CR↑ (confusion rate)	Whether VLM is actively misled toward the target
Q2	Binary: is any text visible in this image?	TVR↓ (text visibility rate)	Whether perturbation conceals textual content
Q3	Two-character forced choice: source char or target char	CA↓ / ASR↑	Direct recognition accuracy (CA = 1− ASR)

Table S5. Full VLM Evaluation Prompts (Q1–Q3). English prompts are used for GPT-5.2 and Gemini 3.0; Chinese prompts are used for GLM-4V to match its primary training distribution. All prompts are deliberately short to suppress chain-of-thought reasoning that could artificially inflate recognition rates. Each call uses `max_tokens`=64 and default temperature.

Q	Metric	English prompt (GPT-5.2, Gemini 3.0)	Chinese prompt (GLM-4V)
Q1	CR↑	*If this image contains a Chinese character, what character is it most likely? Respond with only one Chinese character, or "none" if you cannot identify any character.*	如果图片中包含一个汉字，你认为最可能是哪个字？仅回答一个汉字；如果完全无法识别，回答"无"。
Q2	TVR↓	*Is there a clearly readable Chinese character in this image? Answer only "yes" or "no".*	图片中是否有一个可以清楚辨认的中文字符？仅回答yes或no。
Q3	CA↓ / ASR↑	*Please identify the Chinese character shown in this image. Respond with only the single Chinese character, nothing else.*	请识别图片中显示的中文字符。仅回答一个汉字，不要回答其他内容。

Table S6. Hyperparameters of Benchmarked Adversarial Methods. All methods use author-recommended defaults. ε is reported in the norm space each method natively uses; pixel-space equivalents ($[0, 255]$) are shown in parentheses where they differ.

Method	Norm space	ε	Steps	Surrogate
ASPL	$[-1, 1]$ L_∞	0.05 ($\approx$12.75/255)	200	SD 2.1
Glaze	$[-1, 1]$ L_∞	16/255	300	ViT-L/14 (CLIP)
AMP	$[0, 255]$ L_∞	8/255	300	BLIP / BLIP-2
XTransfer	$[0, 255]$ L_∞	12/255	300	Ensemble $\times$4
Nightshade	$[0, 1]$ L_∞	0.05 ($\approx$12.75/255)	500	SD 2.1
MMCoA	CLIP embedding	1/255	100	ViT-L/14 (CLIP)

Table S7. Confusion Rate CR↑ (%) by VLM and Attack Method (Q1, ID-based Targets). CR measures the fraction of images for which the VLM responds with the *target* character under the most-likely-character prompt. The large GLM-4V values reflect a model-specific tendency to commit to a character guess even under heavy perturbation; GPT-5.2 and Gemini 3.0 more frequently abstain, yielding lower CR alongside near-perfect ASR. **Bold**: best (highest CR) per column.

Method	Gemini 3.0	GPT-5.2	GLM-4V	Avg CR↑
No Attack (baseline)	0.0	0.0	0.0	0.0
MMCoA	0.1	1.0	**92.0**	31.0
AMP	0.0	**1.6**	**98.8**	**33.5**
ASPL	0.0	0.6	97.7	32.8
Nightshade	0.0	0.3	97.3	32.5
XTransfer	0.0	0.4	93.6	31.3
Glaze	**0.2**	0.2	92.6	30.9

Table S8. Text Visibility Rate TVR↓ (%) by VLM and Attack Method (Q2, ID-based Targets). TVR measures the fraction of images for which the VLM reports text as *still visible* after perturbation. Gemini 3.0 exhibits anomalously high TVR even on the clean baseline (98.1%), indicating that it almost always perceives text regardless of perturbation; GPT-5.2 and GLM-4V show much lower post-attack TVR. **Bold**: best (lowest TVR) per column among attacked methods.

Method	Gemini 3.0	GPT-5.2	GLM-4V	Avg TVR↓
No Attack (baseline)	98.1	1.1	0.0	33.1
MMCoA	**97.3**	**3.2**	1.0	**33.8**
AMP	97.0	3.7	**0.0**	**33.6**
ASPL	95.8	7.8	**0.0**	34.5
Nightshade	95.7	10.4	**0.0**	35.4
XTransfer	95.4	10.7	1.0	35.7
Glaze	94.3	12.9	0.6	35.9